Deep Neural Network

Introduction to Convolution Neural Network

Problematique

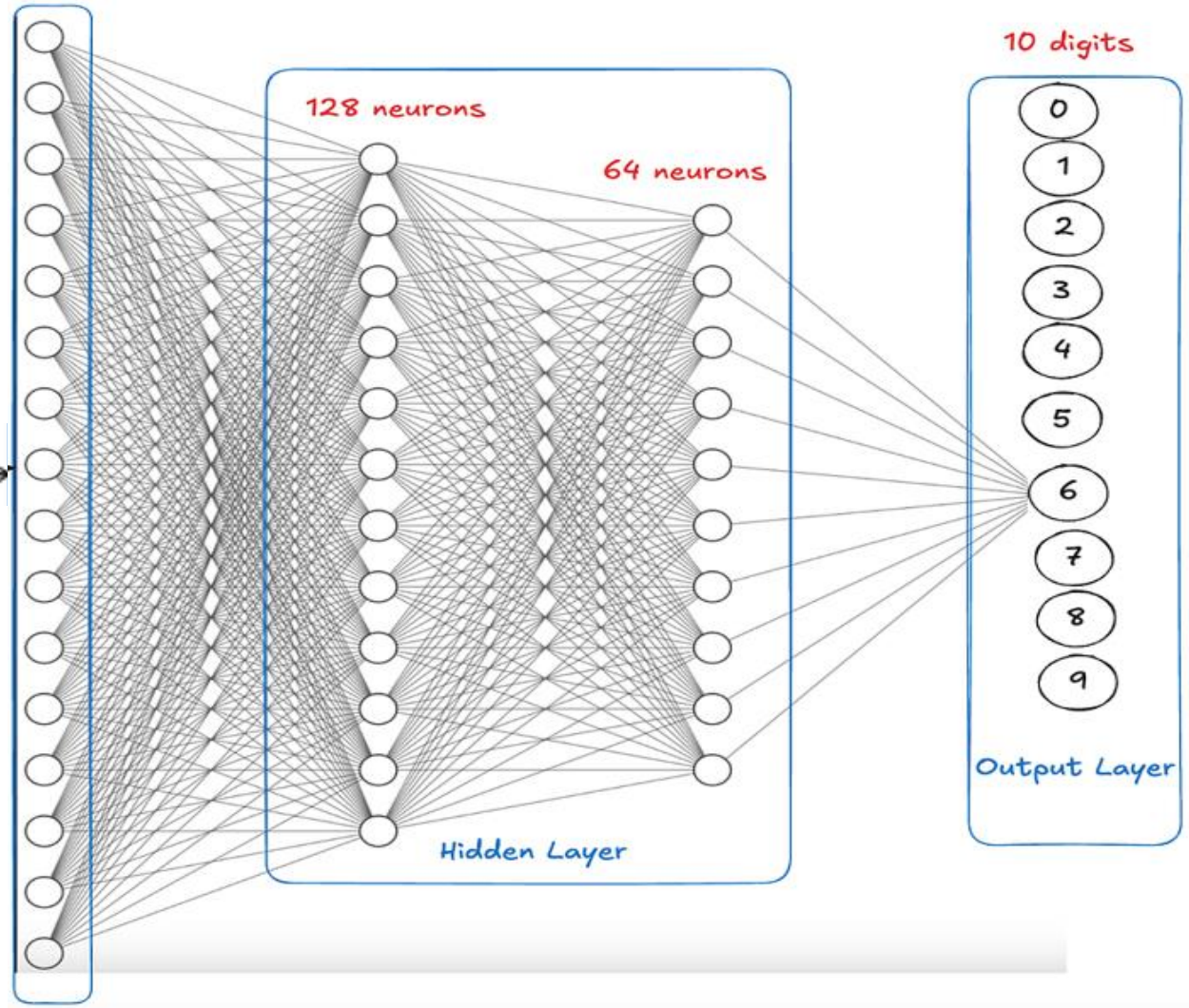
128x128=16384 neurons

128

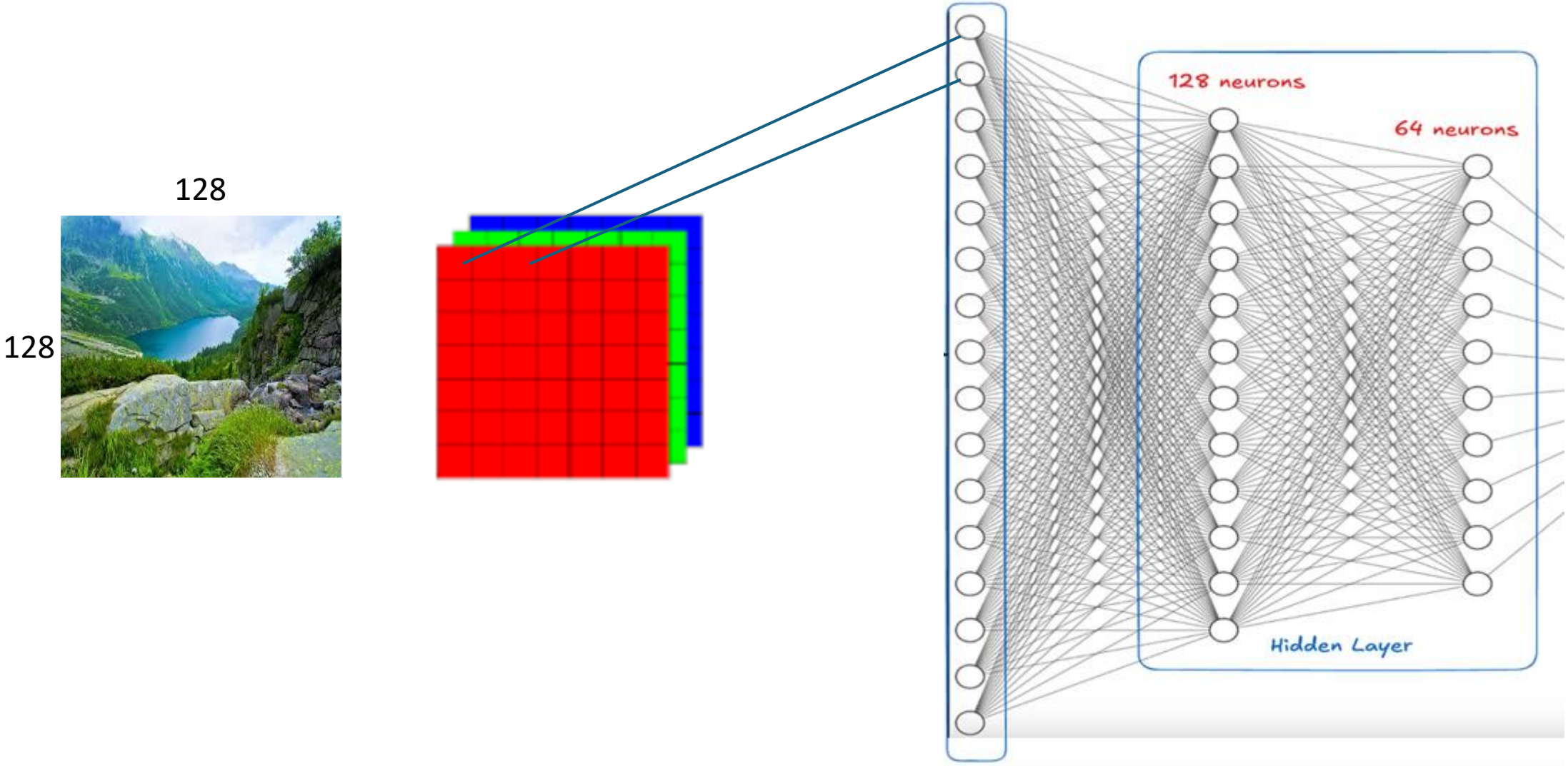
128

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0
0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0
0	0	0	1	1	1	0	0	0	1	1	1	0	0	0	0
0	0	0	1	1	1	0	0	0	1	1	1	0	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0
0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Flaten



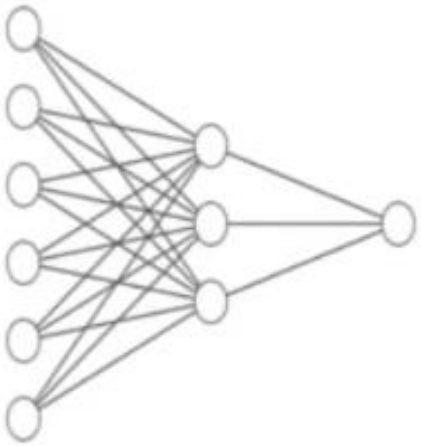
128x128=16384 x128=2 097 152 parameters (weights)



128x128x3=10752 neurons

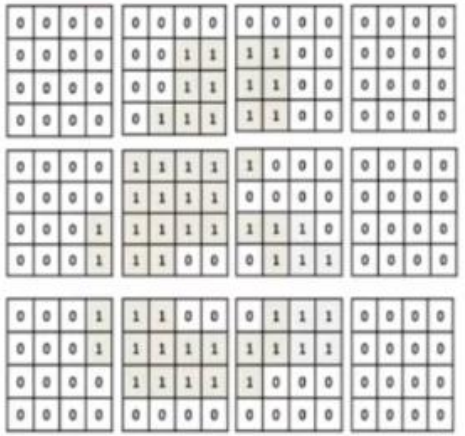
Solution ： Convolutional Neural Network

Convolutional Layer



Filtre or Kernel

1	0	0	1
1	0	0	1
1	0	0	1
1	0	0	1



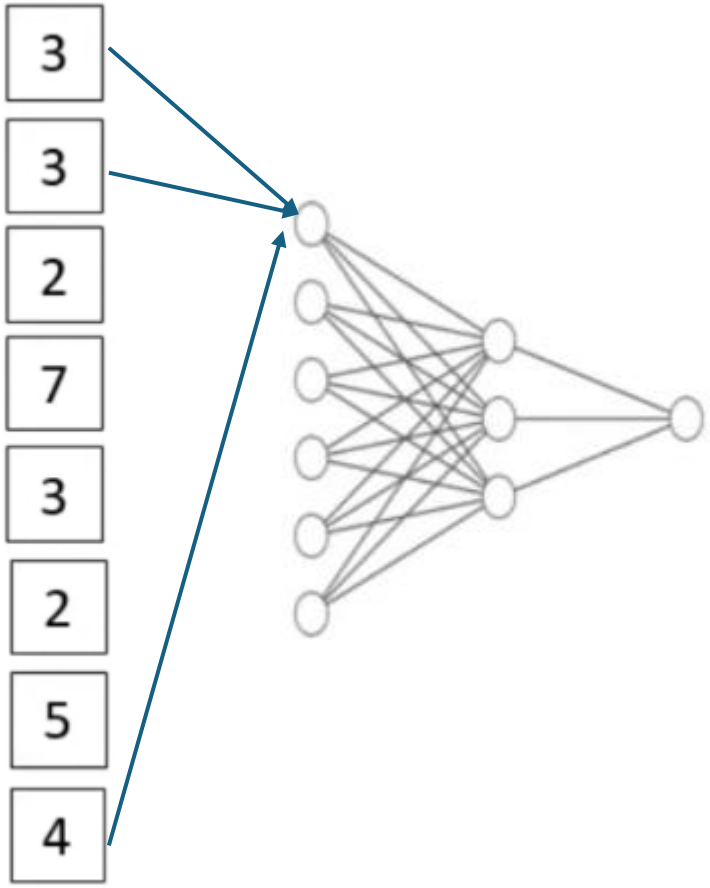
0	3	3	0
2	7	3	0
2	5	4	0

3 Operations : Filtre - Convolution - Polling

Features Detection

Filtre - Convolution - Polling

Convolutional Layer



Convolutional Neural Network (CNN)

$$\Sigma\ 1.w1 + 1.w2 + 1.w3 + 0.w4 + 1.w5 + 1.w6 + 0.w7 + 0.w8 + 1.w9$$

$$\Sigma\ 1.w1 + 1.w2 + 0.w3 + 1.w4 + 1.w5 + 1.w6 + 0.w7 + 1.w8 + 1.w9$$

$$\Sigma\ 1.w1 + 0.w2 + 0.w3 + 1.w4 + 1.w5 + 0.w6 + 1.w7 + 1.w8 + 1.w9$$

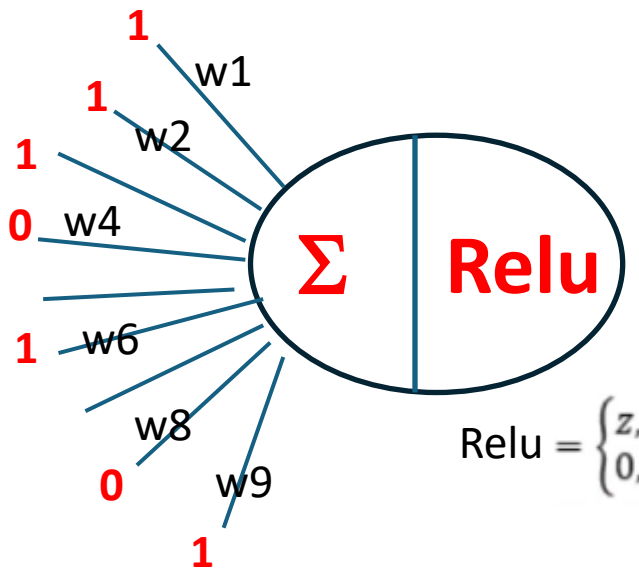
Image

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

$5x5$

Filtre 3x3

W1	W2	W3
W4	W5	W6
W7	W8	W9



$$\text{Relu} = \begin{cases} z, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

Feature Map

F1	F2	F3
F4	F5	F6
F7	F8	F9

MaxPolling

M1	M2
M3	M4

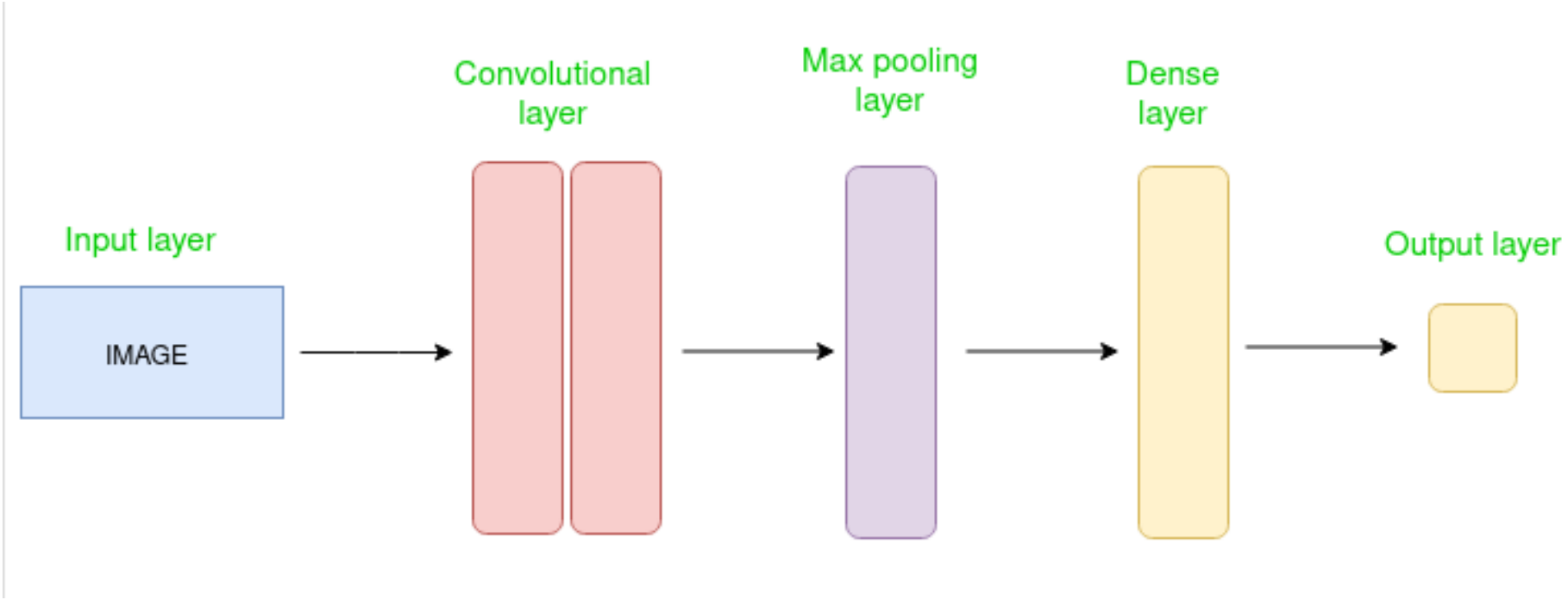
M1
M2
M3
M4

Flatten

Convolutional operation

Stride=1

Introduction to Convolution Neural Network

Convolutional Neural Network (CNN) is an advanced version of artificial neural networks (ANNs), primarily designed to extract features from grid-like matrix datasets. This is particularly useful for visual datasets such as images or videos, where data patterns play a crucial role. CNNs are widely used in computer vision applications due to their effectiveness in processing visual data.

CNNs consist of multiple layers like the input layer, Convolutional layer, pooling layer, and fully connected layers.